

Fab26

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1 INTRODUCTION - THE MOMENT WE ARE IN

The release of ChatGPT in November 2022 has led to a rapid adoption of generative artificial intelligence, particularly LLMs, among the general public **chatgptrelease**. An inevitable consequence of this has been the extended usage of these technologies in the context of education [cite]. Students' usage of LLMs in, for instance, assignments, poses a threat to the traditional model of education, based on the role of the teacher as the center of instruction and memoristic techniques as a demonstration of knowledge acquisition, also known as behaviorism **ertmer1993conductismo**. We observe several reactions among educators, ranging from confusion to prohibition [we could reference the french universities' article here].

Yet these abrupt reactions are not just responses to a new tool; they highlight a long-lasting debate about educational models that the appearance of LLMs has only resurfaced. Theories such as cognitivism, constructivism, and connectivism; which respectively emphasize the construction of knowledge through experience and the formation of learning networks **gortaire2022constructivismo gutierrez2012conectivismo**, have long been challenging the teacher-centered, memoristic model of education. Under these "newer" paradigms of education, the maker community, and fab labs in particular, represent perhaps the most tangible representation of these principles in practice: spaces where learning emerges from making, experimentation, and collaborative problem-solving rather than from passive reception of instructions **gershenfeld2005fab**. In this way, the maker culture does not introduce a new pedagogy, but offers a mature, reference point from which to think about the role that LLMs can play in education going forward.

we choose therefore to see the massive adoption of llms in education (especially among students) not just as a tool change, but as an epistemic change, leaving plenty of room to think beyond isolated tools or "technologies", and onto a root questioning of the educational frameworks that empower learning and making.

[a research question would probably be a good idea here, maybe "What kind of learner do we want to form?" (open enough to leave room for discussion). no. let s put a question like the audio i sent u "what if // why not" based]

<!--The renovation of the way we learn introduced by the maker community, especially when it comes to [] and [] [jorge can say more here. basically the point would be that the maker community is educational avantgarde so from all that has been learnt in this mindset and culture they should contribute to the framing of the role that llms can take in education looking forward.]. we choose therefore to see the massive adoption of llms in education (especially among students) not just as a tool change, but as an epistemic change, leaving plenty of room to think beyond isolated tools or "technologies", and onto a root questioning of the educational frameworks that empower learning and making.-->

2 CONTEXT - TECHNOLOGY & EDUCATION (past, present & future)

2.1 The past - technology as an object of knowledge

In the past, especially based on the nineteenth century model, education was based on a teacher-centered approach. with memorization and reproduction at the core of the aspects valued on a student, knowledge was viewed as something to be "deposited onto" the student [here we can cite the historical study of pedagogical methodologies by the venezuelan researcher on the bibliography] [it could be super cool to make an analogy with industrial production of "educated citizens", in a similar fashion to mass production one-function products on an industrial scale (vs. making and personal fabrication later)]. in this schema, technology has been historically taught as an isolated subject. we're talking about the IT course where you get taught where to click to find a pre-built function in excel as a canonical example of the typical approach taken with regards to technology by this pedagogical model. in this model where the authority of the teacher defines truth, students learn about technology but not how to question it, accepting it as a rigid and functioning black box.

2.2 The present - technology as instrument

<!--[here it would be nice if you wrote a bit more, since you're more into it, jorge. what i would see though would be explaining the following] - learning by doing - fablabs and maker culture in this model, the teacher becomes a facilitator, where technology is a transversal tool in order to make what citizens want to make.

this emphasis here is on "what do i want to make -> how do i make it". it is therefore based in an instrumental mindset when it comes to technology, where we use it and interact with it in a unilateral way.

this is not applicable, however, to LLMs, particularly in education settings, as shown by MIT's studies on the recently coined term "cognitive debt" upon use of these technologies by students [here we should cite it]. we are no longer dealing with a unilateral interaction with technology (one way street), but also with what this technology does to us.

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The late 20th and early 21st centuries brought a shift in pedagogy itself. Where previous models of education had positioned the student as a passive recipient of pre-packaged knowledge, a current of thinkers and practitioners began to insist that people learn best by making, confronting students with real problems and hand them with real tools to iterate towards solutions. The progressive school movement, which started establishing itself during the beginning of the 20th century brought an educational revolution, proposing a pedagogy opposed to the teacher-as-authority model, where the student became the center of the educational process and the teacher recast as facilitator and collaborator, believing that all educational methods should be tested "always opting for the one that best fits depending on the local characteristics, the nature and character of each child" **sola1980educacio**

The core commitments of the progressive school movement have been progressively adopted into contemporary educational frameworks, shifting from knowledge reproduction to competency acquisition to reframe education as not just the accumulation of information, but the development of capacities that transfer across contexts. Learning to know, to do, to be, and to live together **tawil2013** replaced memorization as the measure of an educated person. To develop these competencies it's requires not just reception but practice, as the capacity to do cannot be transmitted through instruction alone, it must be exercised. Under this idea at heart, learning by doing appeared, a model rooted in constructivism and formalized at the 1930s, that proposed that retention and understanding scale directly with experience **rodriguez2014aprender**. In this paradigm where competency is built through action, then the workshop, the laboratory, and the studio are not supplements to

education but its proper site. Under this logic the Fab Labs emerged as spaces that assembled the tools to make that encounter possible, offering the spaces and the digital fabrication tools to let the user thrive through experimentation. What differentiated them from conventional workshops was not the tools available but the pedagogy embedded in their design, in this space digital fabrication empowers users to build upon their own knowledge through projects of their own choosing, where the question driving the learner shifts from what do I need to know? to what do I want to make, and how do I make it?

Under this model though, the relationship between student and technology, is asymmetrical. The student interacts with the technology by bending it, playing with it, iterating toward a desired output, but the technology itself does not push back. It executes, it responds, and resists mainly in physical ways; it does not propose or question. The tools are powerful, but they remain tools as they get shaped by the makers, and it is this assumption that the arrival of large language models in the classroom has begun to question.

2.3 The (emerging) future - technology as a condition of thought

[quite pretentious, but we should think of a nice title for this section probably after we've written more of the paper]

As evident from several research lines such as ... and ... [here we can talk about AI psychosis, sychophancy and self reassuring bubbles or, in learning, cognitive debt], these technologies are not just a tool we use, but they also shape our cognition, [microstate], in the bigger picture [macrostate], social structures. [here the link is very faint, the arguments should be more solid. i think maybe a way could be to get into the epistemological aspect of language and how both are shaped and interconnected, pero igual es demasiado fumada.].

Educational goal must shift from using tools to understanding systems [we can have a deeper look into systems theory, im on it].

This requires a new educational foundation: critical thinking oriented toward technological sovereignty.

3 WHY CRITICAL THINKING BECOMES CENTRAL IN THE AGE OF LLMs

3.1 What is critical thinking?

[a frankenstein definition for our purposes]

synthesized definition + key indicators → ask berta (her research goes very much in that direction)

- reflective judgment: The way individuals understand the nature of knowledge and how they justify beliefs about complex, ill-structured problems. it concerns uncertainty, evidence evaluation, criteria for justification and it is relevant in conditions where problems are open and no single correct answer is available. **KITCHENER198189**

- metacognition: thinking about one's thinking is essential to bring awareness to underlying structures and implicit biases in one's reasoning and understanding.

- another aspect we deem essential surrounding the ability of students (and citizens) to think critically is identifying underlying assumptions. [we should probably explain a bit more. maybe through an analogy or example]

3.2 Critical thinking as a trainable cognitive practice

if we take reflective judgement as defined by kitchener to be a skill parallel to critical thinking, we can see that the findings of this (very relevant) article in the area state that age and educational

level. however, they assert, the development of reflective judgement (which we take as a parallel to critical thinking) is not automatic and education must engage studentss with open-ended questions and ill-structured problems **KITCHENER198189**

exposure to these practices such as metacognition, identifying underlying assumptions in one's (or others') arguments adn statements, we believe, is an effective way to train the "muscle" of critical thinking.

3.3 Why LLMs make critical thinking structurally necessary

LLMs simulate reasoning through very eloquent language. these probabilistic models, far from producing knowledge, produce the most likely word with extreme confidence within their speech. we want to stress the epistemic difference between ideas (which are abstract and can be expressed though different channels, one of them being language) and language itself as an artifact which can very well be empty of ideas or background reasoning, as is the case of LLMs. besides this realization, which is at the very core of our argumentation and non-techno-futuristic interpretation of what LLMs are. one aspect of this, as we mentioned, is the blind trust in eloquent but not "reasoning-backed"-LLMs. we believe, furthermore, that another risk behind using these technologies without critically reflecting on them is passivity. without prior critical capacity, over-reliance on LLMs amplifies passivity and can then lead into a loop of epistemic outsourcing: having less critical thinking skills can lead to over-reliance and thinking outsourcing, which in turn leads to more cognitive debt (see **kosmyna2025brainchatgptaccumulationcognitive**), more passivity and, again, over-reliance and viceversa. "choosing the easy path" can be costly over time, especially when done repeatedly, since we believe friction is an essential aspect of learning and crafting ideas.

4 FROM DEMOCRATIZATION TO TECHNOLOGICAL SOVEREIGNTY

4.1 Limits of "user-friendly democratization"

Nowadays, "Democratizing" is no longer a word which reflects an active participation in building or, in a way, making things beyond just using "user-friendly" platforms. Access to the technologies does not automatically mean understanding or questioning, the "user-friendly" economy is precisely that, <u>user</u> friendly. User being a passive interaction reduced to almost consumption. It contributes to increasing our comfort and dependence on black boxes, and comfort; as nice as it is, does not ask questions.

In Aesop's fable of the fox and the grapes, a fox spots a set of ripe grapes hanging in a branch that is out of his reach. He jumps and jumps, exhausting himself, and cannot get them. Rather than acknowledging his failure and figuring a way up, he walks away muttering: "I can see now plain enough that the grapes are sour, and not fit to be eaten" **godwin1818fables**. This fable represents self-deception, pretending not to want what we were not able to have. Something similar happens with "user-friendly" technology: by removing the effort required to understand how things work, the impulse to try gets removed. Richard Sennett **sennett2019construir** traces this back to Bill Gates' ideal of "friction-free" design, technology so "easy to use" that the user never needs to encounter the machinery beneath. Access widens, but understanding does not. In this scenario, the maker movement represents, in a way, a rejection of this ideal, reclaiming the friction that frictionless design have erased, and with it opening (both critically and literally), modifying, and questioning the tools we depend on. According to Sennett **sennett2019construir**, it is precisely this culture of opening and questioning that sustains critical thinking. A society of users, shielded from complexity by smooth interfaces, loses not just technical literacy but the cognitive habit (no se si esto tiene sentido) of asking what could be otherwise ("what if...?").

Para crear soberanos tecnologicos, tenemos que llevar a cabo una alfabetizacion. No hablamos de como aprender a hacer un powerpoint en una clase de informatica ni de saber cómo usar cada una de

las funciones predefinidas en la UI de chatgpt en el sentido de la educacion de modelo decimononico explicado anteriormente. Nos referimos a una alfabetizacion en el sentido propuesto por paulo Freire (see **freire1998pedagogy**), como una transformacion en el modo en el que ocupamos espacios y que lugar tomamos: el de consumidor industrial pasivo o el de maker. Para esto, creemos que la tecnologia, en tanto que herramienta, deberia estar completamente integrada en curriculums educativos más allá de las delimitaddas clases de nuevas tecnologias o informatica. la ya extendida ia generativa puede ser una oportunidad de cuestionarnos por qué enseñamos los currículums que enseñamos y cómo lo hacemos, y de pensar en nuevas formas actualizadas y empoderadoras de educar. ejemplo de luke davidson enseñando macbeth shakespeare. qué tipo de preguntas podríamos plantearnos en este caso? how can we talk about the big elephant in the room?

4.2 Technological sovereignty

the corruption of the word "democratization" is the reason why we propose a shift towards an educational framework which, instead, places technological sovereignty at the core. The crucial question would then no longer be "how do we democratize technology?"; but rather "how do we create technological sovereigns?". We believe that critical thinking (as explained earlier) through questioning these technologies (implicit assumptions inherently present in their development, who they are built by, or the economic structures that support them, for instance), could be a step towards technological sovereignty.

in a more concrete way, we could make an attempt at a definition of technological sovereignty as the capacity to understand, question, modify and decide over technology, which we consider particularly crucial when it comes to LLMs in education.

4.3 The role of friction in learning

[here @jmuozan can probably say much more] - thinking requires resistance [craft mentality] -> here it could be cool to cite eric sadin's the spectral life book (<https://www.goodreads.com/book/show/211267944-la-vida-espectral>), aunque la cita es de un tal william morris "lo que caracteriza la figura del artesano es que domina el conjunto de las etapas de la profuccion "pudiendo decir que el objeto que creaba era suobra""->

In an educational context dominated by friction-free software, defined by the mass adoption of LLMs, it seems imperative to highlight that it is precisely thinking that requires resistance. The student, just like a craftsman, improves through dwelling in difficulty and lack of understanding. As Richrd Sennett writes in *The Craftsman*, "resistance and ambiguity can be instructive experiences; to work well, every craftsman has to learn from these experiences rather than fight them" **sennett2008craftsman**. The "smart machines" short-circuit the learning process by stepping in where difficulty begins, resolving the struggle before the student has had to inhabit it.

- cognitive effort is pedagogically necessary

" Where is the wisdom we have lost in knowledge? Where is the knowledge we have lost in information? " **eliot1934rock**

- efficiency is not always educationally desirable [we could cite again the MIT cognitive debt paper]

5 EDUCATIONAL FRAMEWORK FOR THE AGE OF LLMS

5.1 What is an educational framework?

A framework should provide a philosophical, epistemological, methodological and analytical lens, through which to examine a topic **grant2014understanding**. Under this paper's conceptualization, we take an educational framework to be the structural foundation which enables and empowers learning. In this sense, it can serve as a "skeleton" of abstract ideas which can then be adapted to

particular cases and contexts (historical, sociological, cultural), as no single framework is necessarily more effective than another, what matters is the fit between a framework and its community of learners. For this reason, understanding education as heavily influenced by the culture and context surrounding the students, shaping how and what they learn as well as how they think and solve problems within their social group **vygotsky1978mind**, we consider an educational framework must be modular, adaptable, and multiscalar in order to serve the multiple local settings of diverse communities.

5.2 Design principles of the proposed framework

An educational framework, then, is not fixed or rigid. It is structured but open, providing a conceptual overview separated from specific methodologies, so that educators, learners, and communities can inhabit it differently depending on their contexts and needs. Much like a cake recipe, the framework captures the essential idea such as the proportions, the process and the intended result, while leaving room for the cook to adapt. A vegan cook may swap cow's milk for soy or almond, while an adventurous one may experiment with different types of chocolate. The recipe should not describe a single cake, as its goal should be to enable many.

[i don't know, these are just some thoughts, feel free to modify. also, they are big statements/principles. i think it's a good idea to just keep them as they are without the explanation after the :, but i left them there in case we want to develop the ideas a bit more, or even place the snippets later in the article]

- critical inquiry before application: what do i want to do, why do i want to use llms? how do i want to use them? [a map] - transparency of cognitive contribution: helps both the educator and the student be more aware and honest with themselves about what areas they have received assistance on - exposure of technological black boxes: information on how they work can contribute to a critical understanding on what tasks we want to use them for and why [@jmuozan can probably say more here], (concept of epistemic opacity / the conference by Títílolá Olojede) - integration across curriculum: llms are already being used by students, let's take the lead on proposing examples of how they can use them to empower their learning, rather than them delegating their thinking on them; not seeing them as an isolated "ai course", but rather integration as a natural tool across curriculum (an example: hamlet and luke) - reflective use of llms: [here i guess it's about asking ourselves what the answers of LLMs are like through critical inquiry. but i don't know how to explain this one in a concise way] - explicit assumptions and bias analysis

5.3 Structure of critical thinking development

[un ejemplo, se puede hacer por ejemplo como un ciclo, it could be so cool to make a diagram like a little tree and then feedbing back. i have it in mind, lets discuss it]

1. Question formulation 2. Information gathering 3. Application 4. Implication analysis 5. Perspective exploration

5.4 Measuring educational impact

A tentative way of evaluating the impact of this approach could include some dimensions rooted in how critical thinking is measured itself: epistemic awareness, reasoning transparency, autonomy in technological decision-system and ability to analyze systems. [comprobar el tema de la madre de nestor]

5.5 Modularity and adaptability

Once again, we would like to stress the importance of modularity and context sensitivity for its adaptation to different educational environments.

6 IMPLEMENTATION IN PRACTICE

6.1 Pedagogical strategies

Even though the framework, as mentioned before, should only give an open recipe for the teacher in order to build the class contents, there are several pedagogical strategies that we have identified through which the framework might take shape in the classroom:

- Opening black boxes: A foundational practice that consists in bringing students into contact with the interior of the tools they use, understanding their applications and implications. This method, though, does not intend the students to become experts; instead, it means offering enough conceptual transparency for the learner to understand the tools and how to use them under their non-frictional interface.

- Documenting thinking processes: In order to mitigate the externalization of the student's own reasoning, making it visible becomes imperative. In order to achieve this, documentation presents a viable way of doing so. Documentation in this context does imply not only the outputs arrived to but the path taken to get there, questions asked along it and the frictions faced as well as the responses done to them.

- Structured transparency in AI use: Alongside the documentation of thinking processes, transparency protocols that reveal the use of LLMs in the different tasks could be of great utility, not only as a way of being "transparent" with the teacher but also as a way of selfreflecting on the usage done, promoting in this case a "honest reflection, responsible use of AI, and clarity in collaborative, tech-assisted creation" **fuentemilla2024ccl**

- assumption mapping [super important, supersuper important and a good way of establishing a first step for metacognitive thinking]

- system analysis tasks [this could be a positive approach in order to enable and continue the metacognitive as an established map of interconnected aspects which play into the task at hand]

It's worth noting that these exposed proposals are just that, proposals, in order to appropriately put them into practice or even discard them in favour of others, they should be evaluated and tested in their respective educational context.

6.2 Learning environments

Independentment de les estratègies adoptades, resulta de gran importància emfatitzar que garantir una correcta i responsable integració dels LLMs a l'aula, implica una participació de tot el professorat. Just as digital literacy was never solely the responsibility of the informatics teacher, nor was writing the exclusive domain of the language teacher, critical engagement with LLMs belongs across the curriculum. A math teacher discussing the reliability of AI generated responses to a math problem, a history teacher questioning the biases in how an LLM frames a historical event, or a language teacher analyzing why chatbots do not construct sentences correctly in catalan language are situations that through transversal implementation trigger critical inquiry.

This distributed model of engagement can be facilitated via the implementation of project-based and service-based learning environments, where boundaries between disciplines get dissolved around a shared problem or goal. When students work towards a real output, whether a designed object or a community intervention, the question of which tools to use, how to use them, and what their limitations arise organically. Academany programs, based on making, documenting, and iterating display this integration of disciplines by requiring students to be able to formulate questions, confront uncertainty, understand tools and reflect on one's own process. In these environments, no single instructor holds authority over the full range of knowledge required to develop one's own final project, the student must navigate disciplines, ask peers, and test tools in order to get to their goal.

6.3 Educating educators

It becomes central for teachers to adopt epistemic rather than instrumental understanding of AI: how can educators empower critical thinking over LLMs if they themselves take a "black box" approach to it? some basic aspect to understand to go beyond digital shintoism (lol inside joke about sadin): (i) statistical aspect of it, (ii) its basic working principles, (iii) whatever else we consider, you know?

7 CASE STUDIES

7.1 FP

Vocational training or Formació Professional (FP) in Catalan/Valencian is a set of educational programs with the goal of enabling individuals to competently practice different professions, providing the necessary training to acquire the professional skills and knowledge specific to each sector. Under this programs, <!--What was implemented--> <!--What worked--> <!--What failed--> <!--What was learned-->

7.2 ONCE

This was a course on generative artificial intelligence for ONCE, which stands for national association of blind people of Spain. Even though originally this association was founded by and for blind people, it has extended to other disabilities. In fact, only one out of the fifteen people who attended the course had some sort of sight limitation (the demographics were very diverse, ranging from diagnosis of squizofrenia, bipolar disorder to chronic fatigue or fibromyalgy). This was a course designed originally to teach digital skills to unemployed people with a disability degree over 33% by the spanish system, in order to increase their possibilities of obtaining an employment position. The learning objectives of the course were provided in advance to the teacher (one of the authors of this article). As mentioned, the contents were already provided and the mere existence of this course as an isolated entity, not incorporated within the curriculum of other courses, goes against the outlook on the role of artificial intelligence on education provided by the authors. We stand for the use of artificial intelligence to be discussed and incorporated in other courses, not to be taught in a stand-alone course. However, we decided to take some of the teaching freedom over the 90 hours of course as an opportunity to experiment with the approach aforementioned. One of the integral parts of the course was active participation and co-design with the students through the exploration of their own ideas and projects as a ground for experimentation. Some of the projects which were brought up by the students within the first sessions included an already-running podcast on the experience of fibromyalgia from the point of view of middle aged women suffering it to the project of museification of a medical center that one of the students had close ties with through historical research of its foundation and bringing the human dimension of medicine to the spotlight. We were thoroughly inspired by the learn-by-doing and maker spaces philosophy, which is why students were encouraged to be creative on the challenges they brought to the table and they would like to work on during the course as a testing grounds for the capabilities of generative artificial intelligence. It must be noted that the students were hardly acquainted with these technologies; which posed on one hand a challenge and, on the other, an opportunity to define together their outlook on them from a critical thinking lens.

Students would receive a short introduction on a technical aspect, for instance around a question like "what's behind a prompt?" as question formulation (see Structure of Critical Thinking Formulation above). This question in particular was formulated as proxy for opening black boxes in particular. As a way to warm up (as well as for the teacher to be aware of the starting point of students), students would be asked to explain briefly through an interactive quiz on the screen. The students seemed to be very motivated by using playful methods (like quizzes or roulettes to see who would present next, for instance). Once their preliminary answers were in screen, we would comment a bit on them. They would be asked to volunteer to explain their answers if they felt comfortable

with it. This served as a very solid ground to have the student's active participation and attention in order to start the theoretical explanation (for instance on how models are trained with large amounts of uncensored internet data or general analogies on how images are generated in the case of multimodal AI) (see Information in the diagram on critical thinking). Students were encouraged then to, around their project, be creative incorporating the new knowledge in an additional feature. For instance, when it comes to the fibromyalgia podcast, one of the students decided to formulate a prompt to create a logo for their project (see Applications in the diagram on critical thinking). Lots of times the result provided by multimodal AI would not be as students expected. They would be asked to explain what they thought the reason for that was, given the information they had learned previously (in the case of the logo for the podcast, for instance, the student mentioned the need to associate language with image. She discussed the limitations of both her knowledge on "the language of images", but also the fact that, since they are two distinct media, to begin with, it could be that it is hard for the model to learn how to connect both). This served as a great example on how applications and experimentation created the grounds to discuss implications and different perspectives on the limitations of these technologies, in this case. This had naturally created a new question formulation in order to continue critical inquiry following the curiosity of students.